

HydroShear: Original Implementation and Our UV-Metric Implementation

1 Notation

Let $\mathbf{p}_i \in \mathbb{R}^3$ be the i -th marker point on the elastomer surface, and let

$$\mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \in \mathbb{R}^2 \quad (1)$$

be its tactile image coordinate. HydroShear decomposes the marker displacement field into dilation and shear:

$$\mathbf{M}_t(\mathbf{x}, {}^E\mathbf{X}_{0:t}^I) = \mathbf{M}_t^d(\mathbf{x}) + \mathbf{M}_t^s(\mathbf{x}, {}^E\mathbf{X}_{0:t}^I). \quad (2)$$

The original paper writes the tactile output as a 2D marker-flow field. The local HydroShear code, however, computes an intermediate 3D vector field and then projects or slices it to obtain a 2D image-space marker flow.

2 Original HydroShear Implementation

2.1 Original Dilation

Let $\phi_I(\cdot)$ be the signed distance function (SDF) of the indenter. A tactile marker/source point is in contact with the indenter if $\phi_I(\mathbf{p}_k) < 0$. Define

$$C_t = \{k \mid \phi_I(\mathbf{p}_k) < 0\}, \quad (3)$$

$$h_k^d = -\phi_I(\mathbf{p}_k). \quad (4)$$

At the paper-formula level, the dilation displacement is written on the 2D tactile plane:

$$\mathbf{v}_{ik} = \mathbf{x}_i - \mathbf{x}_k, \quad (5)$$

$$\mathbf{M}_{i,t}^d = c_d \sum_{k \in C_t} h_k^d \mathbf{v}_{ik} \exp\{-\lambda_d \|\mathbf{v}_{ik}\|_2^2\}. \quad (6)$$

The local original implementation uses the same Gaussian idea but computes the intermediate displacement in 3D:

$$\mathbf{r}_{ik}^{3D} = \mathbf{p}_i - \mathbf{p}_k, \quad (7)$$

$$\mathbf{M}_{i,t}^{d,3D} = c_d \sum_{k \in C_t} h_k^d \mathbf{r}_{ik}^{3D} \exp\{-\lambda_d \|\mathbf{r}_{ik}^{3D}\|_2^2\}. \quad (8)$$

2.2 Original Hydroelastic Force Tracker

Let $\mathbf{o}_{j,t} \in \mathbb{R}^3$ be the j -th indenter surface sample in the elastomer frame at time t . Let $\phi_E(\cdot)$ be the elastomer SDF. The original tracker first estimates how much of the object-point motion

occurs inside the elastomer:

$$\alpha_{j,t} = \frac{\text{ReLU}(-\phi_E(\mathbf{o}_{j,t})) - \text{ReLU}(-\phi_E(\mathbf{o}_{j,t-1}))}{\phi_E(\mathbf{o}_{j,t}) - \phi_E(\mathbf{o}_{j,t-1}) + \epsilon}, \quad (9)$$

$$\mathbf{d}_{j,t} = \alpha_{j,t} (\mathbf{o}_{j,t} - \mathbf{o}_{j,t-1}). \quad (10)$$

With contact normal $\mathbf{n}_j$, the in-contact displacement is split into normal and tangential components:

$$d_{j,t}^n = \mathbf{d}_{j,t}^T \mathbf{n}_j, \quad (11)$$

$$\mathbf{d}_{j,t}^\tau = \mathbf{d}_{j,t} - d_{j,t}^n \mathbf{n}_j. \quad (12)$$

The accumulated normal and tangential quantities are then

$$f_{j,t}^n = \tilde{f}_{j,t-1}^n + E A_j d_{j,t}^n, \quad (13)$$

$$\mathbf{f}_{j,t}^\tau = \tilde{\mathbf{f}}_{j,t-1}^\tau + K A_j \mathbf{d}_{j,t}^\tau. \quad (14)$$

Normal pulling is removed, and the tangential component is clipped by Coulomb friction:

$$\bar{f}_{j,t}^n = \text{ReLU}(f_{j,t}^n), \quad (15)$$

$$\bar{\mathbf{f}}_{j,t}^\tau = \min \left(1, \frac{\mu \bar{f}_{j,t}^n}{\|\mathbf{f}_{j,t}^\tau\|_2 + \epsilon} \right) \mathbf{f}_{j,t}^\tau. \quad (16)$$

Finally, the force-like tracked vector is reset when the object sample is not inside the elastomer:

$$\tilde{\mathbf{f}}_{j,t} = H(-\phi_E(\mathbf{o}_{j,t})) (\bar{f}_{j,t}^n \mathbf{n}_j + \bar{\mathbf{f}}_{j,t}^\tau). \quad (17)$$

2.3 Original Projection Point

The paper distinguishes the contact force $\tilde{\mathbf{f}}_{j,t}$ from a projection displacement $\hat{\mathbf{f}}_{j,t}$. The projected elastomer-side contact point is

$$\hat{\mathbf{o}}_{j,t} = \mathbf{o}_{j,t} + \hat{\mathbf{f}}_{j,t}. \quad (18)$$

Under the simplifying assumption $E = K$, $\mu = \hat{\mu}$, and uniform area $A_j = A$, the paper relates the force and projection displacement by

$$\tilde{\mathbf{f}}_{j,t} = K A \hat{\mathbf{f}}_{j,t}. \quad (19)$$

This distinction is important because a physical force should not be directly added to a position. In code, the calibrated tracked vector is often used as a displacement-like quantity for shifting the Gaussian center.

2.4 Original Shear

Let

$$K_t = \{j \mid \phi_E(\mathbf{o}_{j,t}) < 0\} \quad (20)$$

be the set of in-contact indenter surface samples. In the paper-level 2D formula, shear is aggregated from the projected contact location $\hat{\mathbf{o}}_{j,t}^{xy}$:

$$h_j^s = -\phi_E(\mathbf{o}_{j,t}), \quad (21)$$

$$\mathbf{w}_{ij} = \mathbf{x}_i - \hat{\mathbf{o}}_{j,t}^{xy}, \quad (22)$$

$$\mathbf{M}_{i,t}^s = c_s \sum_{j \in K_t} h_j^s \left(-\tilde{\mathbf{f}}_{j,t}^{xy} \right) \exp\{-\lambda_s \|\mathbf{w}_{ij}\|_2^2\}. \quad (23)$$

The local original implementation can be viewed as a 3D version. Let $\mathbf{b}_{j,t}$ denote the calibrated displacement-like tracked vector used by the code. The Gaussian source center is

$$\mathbf{a}_{j,t} = \mathbf{o}_{j,t} + \mathbf{b}_{j,t}. \quad (24)$$

Then the implemented 3D shear aggregation is

$$\mathbf{M}_{i,t}^{s,3D} = c_s \sum_{j \in K_t} h_j^s(-\mathbf{b}_{j,t}) \exp\{-\lambda_s \|\mathbf{p}_i - \mathbf{a}_{j,t}\|_2^2\}. \quad (25)$$

The original local code then forms

$$\mathbf{M}_{i,t}^{3D} = \mathbf{M}_{i,t}^{d,3D} + \mathbf{M}_{i,t}^{s,3D}. \quad (26)$$

3 Our Current Implementation

3.1 Raycaster Coordinates and Metric

For the curved Revo fingertip, every marker or object sample is associated with a raycaster grid coordinate

$$\mathbf{u}_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix} \in \mathbb{R}^2, \quad (27)$$

where u is the grid-column coordinate and v is the grid-row coordinate. Let $\mathbf{g}_u, \mathbf{g}_v \in \mathbb{R}^3$ be the world-space metric vectors corresponding to a one-pixel step along the raycaster column and row directions. In the current implementation these are global per-frame metric vectors estimated from the ray grid:

$$J = [\mathbf{g}_u \quad \mathbf{g}_v] \in \mathbb{R}^{3 \times 2}, \quad G = J^T J \in \mathbb{R}^{2 \times 2}. \quad (28)$$

For two ray-grid points $\mathbf{u}_a$ and $\mathbf{u}_b$,

$$\Delta \mathbf{u}_{ab} = \mathbf{u}_a - \mathbf{u}_b, \quad (29)$$

$$\mathbf{r}_{ab}^{uv} = J \Delta \mathbf{u}_{ab}, \quad (30)$$

$$d_{uv}^2(a, b) = \|\mathbf{r}_{ab}^{uv}\|_2^2 = \Delta \mathbf{u}_{ab}^T G \Delta \mathbf{u}_{ab}. \quad (31)$$

This is a raycaster UV metric distance. It is more surface-aware than directly using the 3D chord distance $\|\mathbf{p}_i - \mathbf{p}_j\|_2$, but it is still not a full geodesic shortest-path distance over the mesh.

3.2 Our Dilation

In the current implementation, contact/penetration for marker source points is estimated from the ray-depth field. Let h_k^{uv} be this depth-based penetration estimate. Our dilation field is

$$\mathbf{M}_{i,t}^{d,uv} = c_d \sum_{k \in C_t} h_k^{uv} \mathbf{r}_{ik}^{uv} \exp\{-\lambda_d d_{uv}^2(i, k)\}. \quad (32)$$

Compared with Eq. (8), the Gaussian vector and distance are no longer computed by the direct 3D difference $\mathbf{p}_i - \mathbf{p}_k$. They are computed through the raycaster UV metric in Eq. (31).

3.3 Our Force Split

The recursive tracker still produces a per-object-sample tracked vector $\mathbf{f}_{j,t} \in \mathbb{R}^3$. Let $\mathbf{n}_j$ be the contact normal used for splitting this vector. We compute

$$q_j = \mathbf{f}_{j,t}^T \mathbf{n}_j, \quad (33)$$

$$h_j^{uv} = \max(q_j, 0), \quad (34)$$

$$\mathbf{f}_{j,t}^\tau = \mathbf{f}_{j,t} - q_j \mathbf{n}_j. \quad (35)$$

Here the normal component q_j is used as the pressure/height weight, while the tangential component $\mathbf{f}_{j,t}^\tau$ is used to shift and propagate the shear field.

3.4 Our UV Shear Center

The current implementation moves the shear Gaussian center in raycaster UV space using only the tangential component:

$$\Delta \mathbf{u}_j^\tau = J^+ \mathbf{f}_{j,t}^\tau, \quad (36)$$

$$\hat{\mathbf{u}}_{j,t} = \mathbf{u}_{j,t} + \Delta \mathbf{u}_j^\tau. \quad (37)$$

Here J^+ is the pseudoinverse of the raycaster metric basis. This maps a world-space tangential vector back to a 2D UV displacement.

3.5 Our Tangent-Plane Vector Projection

Before the source shear vector from object sample j contributes to marker i , it is projected onto marker i 's tangent plane:

$$P_i = I - \mathbf{n}_i \mathbf{n}_i^T, \quad (38)$$

$$T_{j \rightarrow i}(\mathbf{f}_{j,t}^\tau) \approx P_i \mathbf{f}_{j,t}^\tau. \quad (39)$$

This is an approximate tangent-plane projection. It is not a full parallel transport operator along a geodesic.

3.6 Our Shear

The final UV-metric shear field is

$$\mathbf{M}_{i,t}^{s,uv} = c_s \sum_{j \in K_t} h_j^{uv} (-P_i \mathbf{f}_{j,t}^\tau) \exp\{-\lambda_s d_{uv}^2(i, \hat{\mathbf{u}}_{j,t})\}. \quad (40)$$

The current total 3D marker displacement is

$$\mathbf{M}_{i,t}^{uv} = \mathbf{M}_{i,t}^{d,uv} + \mathbf{M}_{i,t}^{s,uv} \in \mathbb{R}^3. \quad (41)$$

3.7 Our Visualization Projection

For visualization, the 3D displacement is projected onto the normalized raycaster axes:

$$\Delta u_i = s_{\text{px}} \mathbf{M}_{i,t}^{uv} \cdot \hat{\mathbf{g}}_u, \quad (42)$$

$$\Delta v_i = s_{\text{px}} \mathbf{M}_{i,t}^{uv} \cdot \hat{\mathbf{g}}_v, \quad (43)$$

where

$$\hat{\mathbf{g}}_u = \frac{\mathbf{g}_u}{\|\mathbf{g}_u\|_2 + \epsilon}, \quad \hat{\mathbf{g}}_v = \frac{\mathbf{g}_v}{\|\mathbf{g}_v\|_2 + \epsilon}, \quad (44)$$

and s_{px} is the pixel-per-meter arrow scale.

4 What We Changed

4.1 Change 1: Euclidean 3D Kernel to Raycaster UV-Metric Kernel

The original local implementation spreads dilation and shear with a direct 3D Euclidean distance:

$$\|\mathbf{p}_i - \mathbf{p}_j\|_2^2. \quad (45)$$

Our implementation replaces this with the raycaster UV metric:

$$d_{uv}^2(i, j) = (\mathbf{u}_i - \mathbf{u}_j)^T G(\mathbf{u}_i - \mathbf{u}_j). \quad (46)$$

This makes the Gaussian propagation follow the ray-grid parameterization of the curved fingertip instead of the straight 3D chord between two points.

4.2 Change 2: Tangential Shift of the Shear Center

The original code-level implementation shifts the shear Gaussian center with the full tracked vector:

$$\mathbf{a}_{j,t} = \mathbf{o}_{j,t} + \mathbf{b}_{j,t}. \quad (47)$$

Our implementation first removes the normal component:

$$\mathbf{f}_{j,t}^\tau = \mathbf{f}_{j,t} - (\mathbf{f}_{j,t}^T \mathbf{n}_j) \mathbf{n}_j, \quad (48)$$

and then shifts the center in UV space:

$$\hat{\mathbf{u}}_{j,t} = \mathbf{u}_{j,t} + J^+ \mathbf{f}_{j,t}^\tau. \quad (49)$$

Therefore, pressing into the sensor mostly changes the pressure weight, while tangential motion shifts the shear source center.

4.3 Change 3: Direct Vector Sum to Marker Tangent-Plane Projection

The original code-level shear contribution uses the same source vector for all markers:

$$-\mathbf{b}_{j,t}. \quad (50)$$

Our implementation projects the source tangential vector onto each target marker's tangent plane:

$$-P_i \mathbf{f}_{j,t}^\tau = -(I - \mathbf{n}_i \mathbf{n}_i^T) \mathbf{f}_{j,t}^\tau. \quad (51)$$

This makes the propagated vector compatible with each marker's local surface tangent plane.

4.4 Change 4: Raycaster-Axis Visualization

The original implementation ultimately reads the tactile field in a 2D sensor plane. Our curved-fingertip implementation first computes a 3D displacement on the surface and then projects it to the raycaster image axes:

$$(\Delta u_i, \Delta v_i) = s_{\text{px}} (\mathbf{M}_{i,t}^{uv} \cdot \hat{\mathbf{g}}_u, \mathbf{M}_{i,t}^{uv} \cdot \hat{\mathbf{g}}_v). \quad (52)$$

4.5 Current Limitations

The current UV metric is still an approximation. It uses global per-frame metric vectors $\mathbf{g}_u$ and $\mathbf{g}_v$, so it is not a fully local surface metric $G(\mathbf{u})$. It also does not compute a true mesh geodesic shortest path. In addition, the RevoLab implementation estimates contact depth from the ray/grid surface representation instead of querying a true volumetric elastomer mesh SDF.

5 Compact Comparison

Component	Original implementation	Our implementation
Dilation distance	$\ \mathbf{p}_i - \mathbf{p}_k\ _2^2$	$d_{uv}^2(i, k) = \Delta \mathbf{u}^T G \Delta \mathbf{u}$
Shear center	$\mathbf{o}_{j,t} + \mathbf{b}_{j,t}$	$\mathbf{u}_{j,t} + J^+ \mathbf{f}_{j,t}^\tau$
Normal component	Can enter the full tracked vector	Used as pressure weight h_j^{uv}
Shear vector	$-\mathbf{b}_{j,t}$	$-P_i \mathbf{f}_{j,t}^\tau$
Surface handling	Direct 3D Euclidean propagation	Raycaster UV-metric propagation
Visualization	Planar tactile/image coordinates	Projection onto raycaster axes